



ME- 310

Design of Machine Elements

4 DOF-Scara

Group 5

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ABSTRACT

This project focuses on the design, analysis, and development of a SCARA (Selective Compliance Assembly Robot Arm) robot for industrial pick-and-place applications. The robot was designed using CAD software and analyzed under different loading conditions to evaluate its structural strength and operational performance. Static analysis, dynamic analysis, force analysis, bending moment calculations, critical stress calculations, and fatigue failure analysis were performed to ensure safe operation of the mechanism. Different hardware components such as servo motors, controller boards, bearings, shafts, and link mechanisms were integrated to develop the complete robotic system. The proposed design provides advantages such as compact structure, reduced weight, improved motion accuracy, and ease of operation. The project demonstrates the practical implementation of robotic automation for lightweight industrial applications.

INTRODUCTION

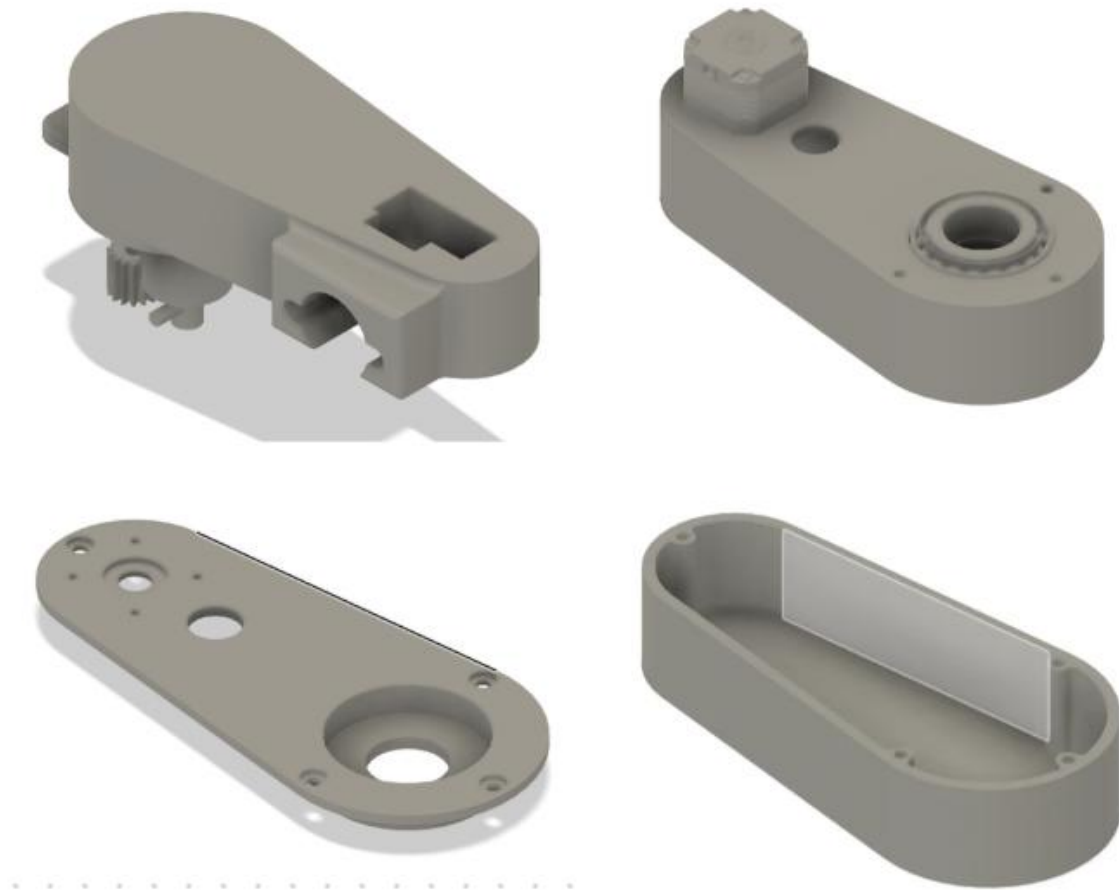
Industrial automation has become an important requirement in modern manufacturing industries due to the increasing demand for productivity, precision, and reduced human effort. Robotic systems are widely used for repetitive tasks such as pick-and-place operations, assembly, packaging, and material handling. However, many industrial robotic systems are expensive and complex for small-scale applications.

The main motivation behind this project is to develop a compact, economical, and efficient SCARA robot capable of performing automated operations with good positional accuracy and structural stability. The SCARA mechanism was selected because of its high-speed motion, simple construction, easy control, and suitability for industrial automation applications.

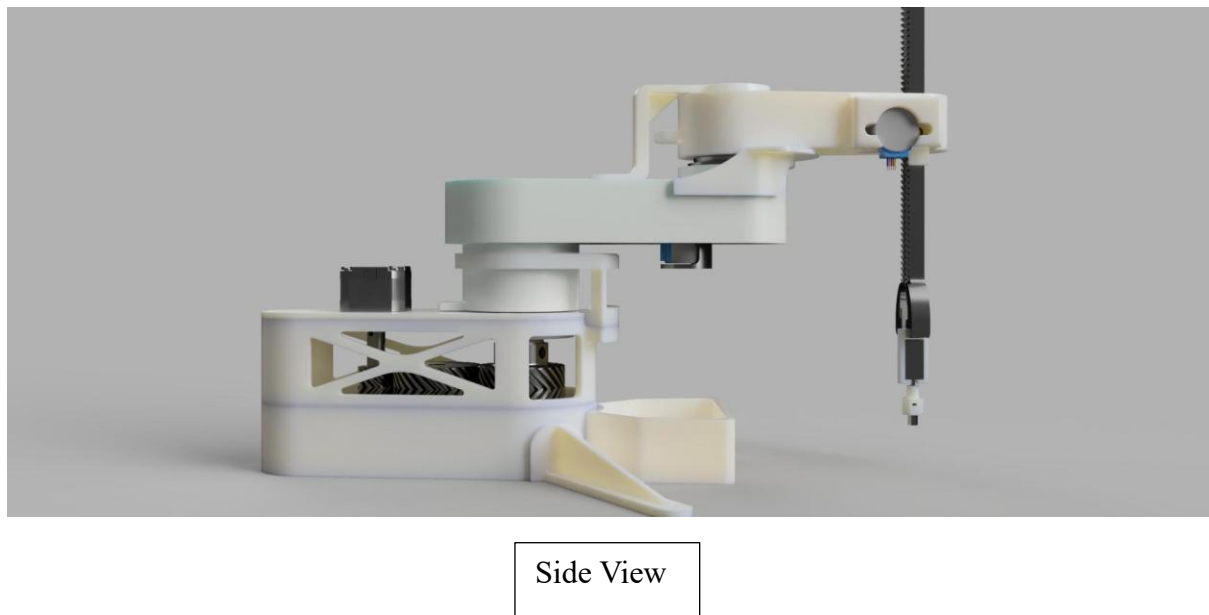
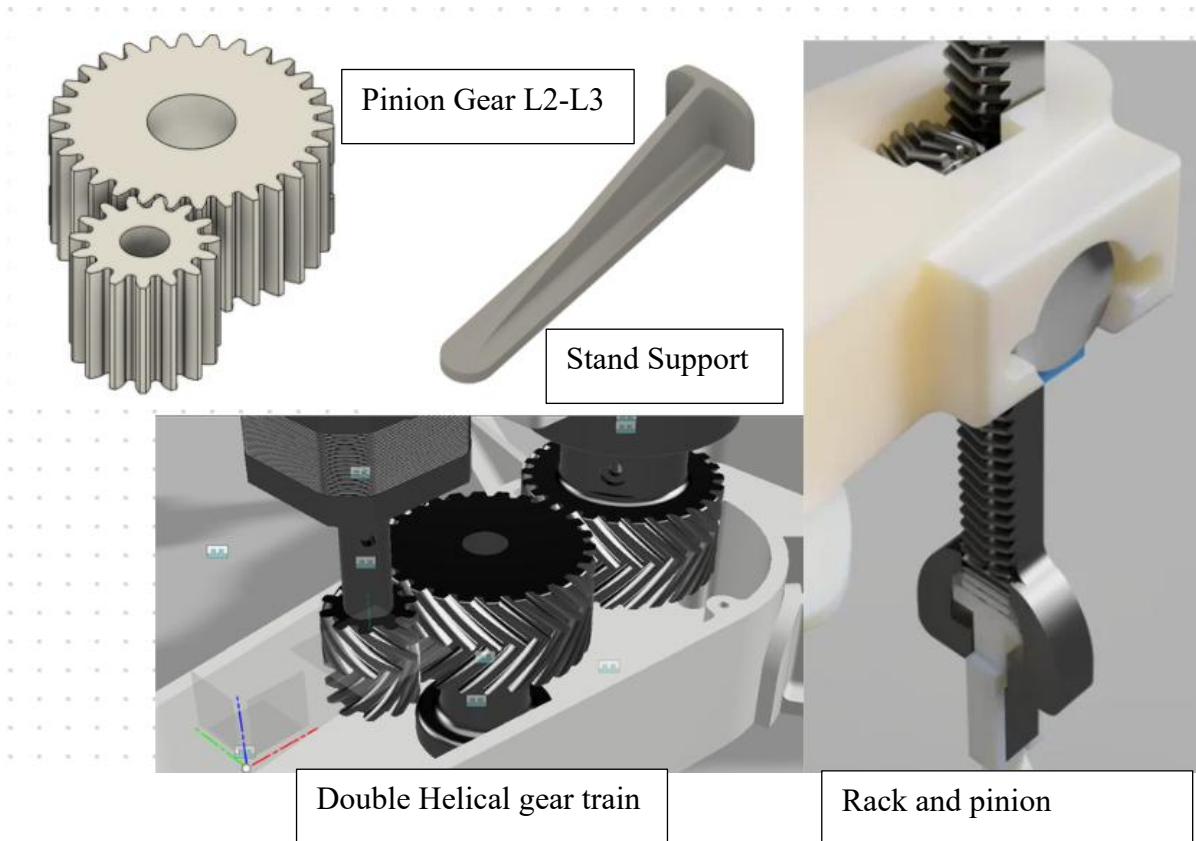
The project also aims to understand practical engineering concepts such as mechanism design, force analysis, stress analysis, fatigue failure, and system integration. The developed system provides an educational and industrial platform for studying robotic automation and mechanical system design.

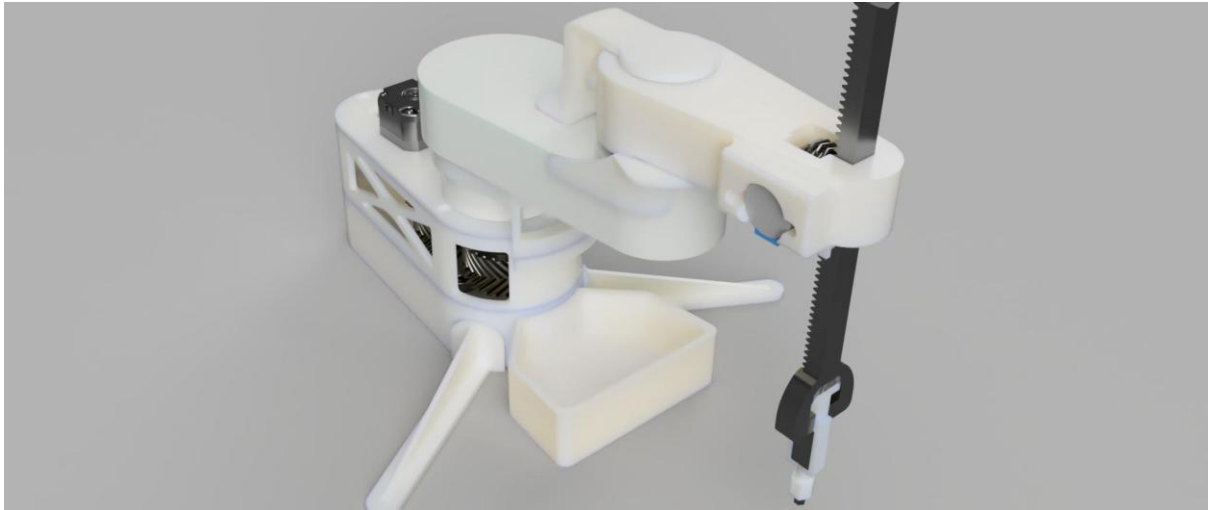
CAD DESIGN

The complete SCARA robot was designed using CAD software to visualize the structure, dimensions, assembly, and working mechanism before fabrication. Different parts such as the base frame, rotating arms, motor mounts, shafts, and end effector were modeled separately and assembled to verify proper motion and alignment.

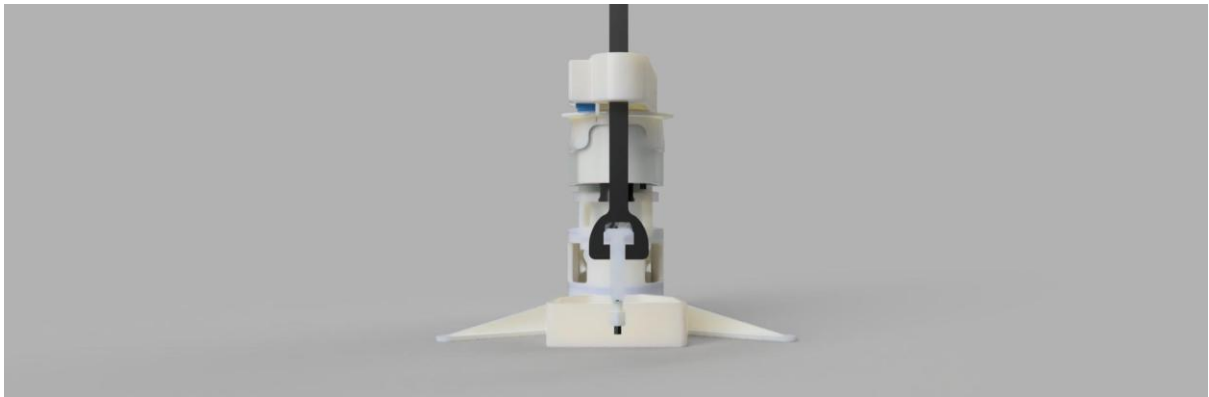


Different Components





Isometric View



Front View

DIMENSIONS

Component	Symbol	Dimension/Value	Unit
Link 1 Length	(L1)	140	mm
Link 2 Length	(L2)	140	mm
Link 3 Length	(L3)	120	mm
Rack working height	(L4)	60	mm
Link 1 Mass	(m1)	1197	g
Link 2 Mass	(m2)	740	g
Link 3 Mass	(m3)	330+140	g
Total Arm Reach	(L)	260	mm
Center of Mass of Link 1		(-404,0,8)	mm
Center of Mass of Link 2		(79,0,74)	mm
Center of Mass of Link 3		(215,0,122)	mm

LINK EXPLANATION

Link 1

Link 1 is the base has a length of 140 mm. It acts as the primary supporting arm of the SCARA mechanism and transfers motion from the motor to the remaining links. The mass of Link 1 is approximately 1197 g.

Link 2

Link 2 is connected to Link 1 through revolute Joint J2 and also has a length of 140 mm. This link provides additional reach and flexibility to the robotic arm. The mass of Link 2 is approximately 740 g.

Link 3

Link 3 is the end link of the mechanism with a length of 120 mm. It carries the end effector and payload during operation. The total mass acting on Link 3 is approximately 330+140.

As our desired workspace has an extent of 260 mm, therefore chosen the dimension of link2 and link3 as 140 and 120 respectively. For link 1,reference from prior design is taken by taking the same ratio of links.

Joint Mechanism

The robotic arm consists of two revolute joints:

- J1: Base rotational joint
- J2: Intermediate rotational joint
- J3: Rack and pinion between link 3 and 4
- J4: Rotational motion of geared motor at end effector

The links were designed to provide sufficient stiffness and strength while maintaining smooth motion and reduced vibration during operation.

Kinematics equations (DH Table):

$${}^0T_1 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 140 \cos \theta_1 \\ \sin \theta_1 & \cos \theta_1 & 0 & 140 \sin \theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1T_2 = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & 120 \cos \theta_2 \\ \sin \theta_2 & \cos \theta_2 & 0 & 120 \sin \theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2T_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_3 = {}^0T_1 {}^1T_2 {}^2T_3$$

$${}^0T_3 = \begin{bmatrix} c_{12} & -s_{12} & 0 & x \\ s_{12} & c_{12} & 0 & y \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$c_{12} = \cos(\theta_1 + \theta_2)$$

$$s_{12} = \sin(\theta_1 + \theta_2)$$

$$x = 140 \cos \theta_1 + 120 \cos(\theta_1 + \theta_2)$$

$$y = 140 \sin \theta_1 + 120 \sin(\theta_1 + \theta_2)$$

$$z = d_3$$

Centre of Gravity of each part

Link	Mass	x	Y	Z
1	1196.99	-44	0	8
2	741.533	79	0	74
3	189.823	214	0	122
4	140.473	216	0	123

- Taking lower bearing centre as origin

Link 1 (Base Link):

- Fixed base structure
- Houses motor for first revolute joint (R_1)

Link 2 (First Arm):

- Rotates about base (R_1)
- Provides primary planar motion

Link 3 (Second Arm):

- Rotates relative to Link 2 (R_2)
- Extends reach in XY plane

Link 4 (End-Effector Assembly):

- Includes:
 - Prismatic joint (Z translation)
 - Rotational wrist (R_4)
 - Gripper / tool (for screw handling)

Stability against Toppling

- Total Mass = 2.268 kg

- Total Weight (W) = mg
 $= 2.268 \times 9.81$
 $= 22.257 \text{ N}$

Using $\Sigma F_y = 0$,

- We get Normal Reaction = W
 $= 22.257 \text{ N}$

Final COG Location for soft home position:

$$X = \frac{\Sigma (m \times x)}{\Sigma m}$$

$$= 33.296 \text{ mm}$$

$$Y = \frac{\Sigma (m \times y)}{\Sigma m}$$

$$= 0 \text{ mm}$$

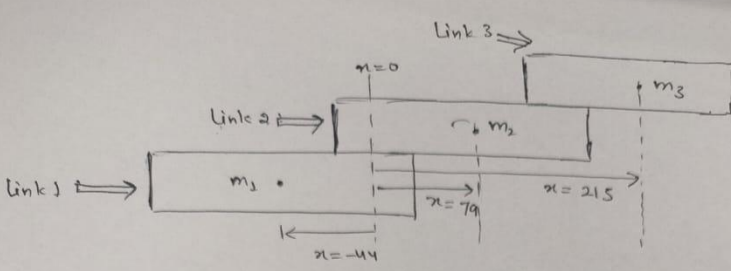
Stability Check

- The COG lies within the base support region
 $X_{\text{base, extremes}} = \{-194.437, 136.133\}$
- Hence, no toppling under static condition

LOADING ASSUMPTIONS:

STATIC ANALYSIS:

Static analysis was performed to determine the deformation and equivalent stress developed in the robotic structure under stationary loading conditions.



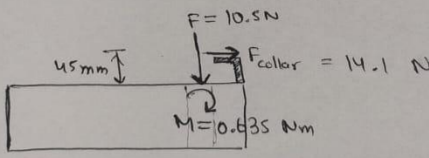
$m_1 = 1197 \text{ g}$
 $m_2 = 741.5 \text{ g}$
 $m_3 = 330.3 \text{ g}$

moment due to m_2 & m_3 on link 1 (at $x=0$) is :

$$= (m_2 \times 79 + m_3 \times 215) \times 9.81$$
$$= 1.27 \text{ Nm}$$

weight of m_2 and m_3 acting on link 1 $= (m_2 + m_3)g$
 $= 10.5 \text{ N}$

→ Consider collar as part of link 1.

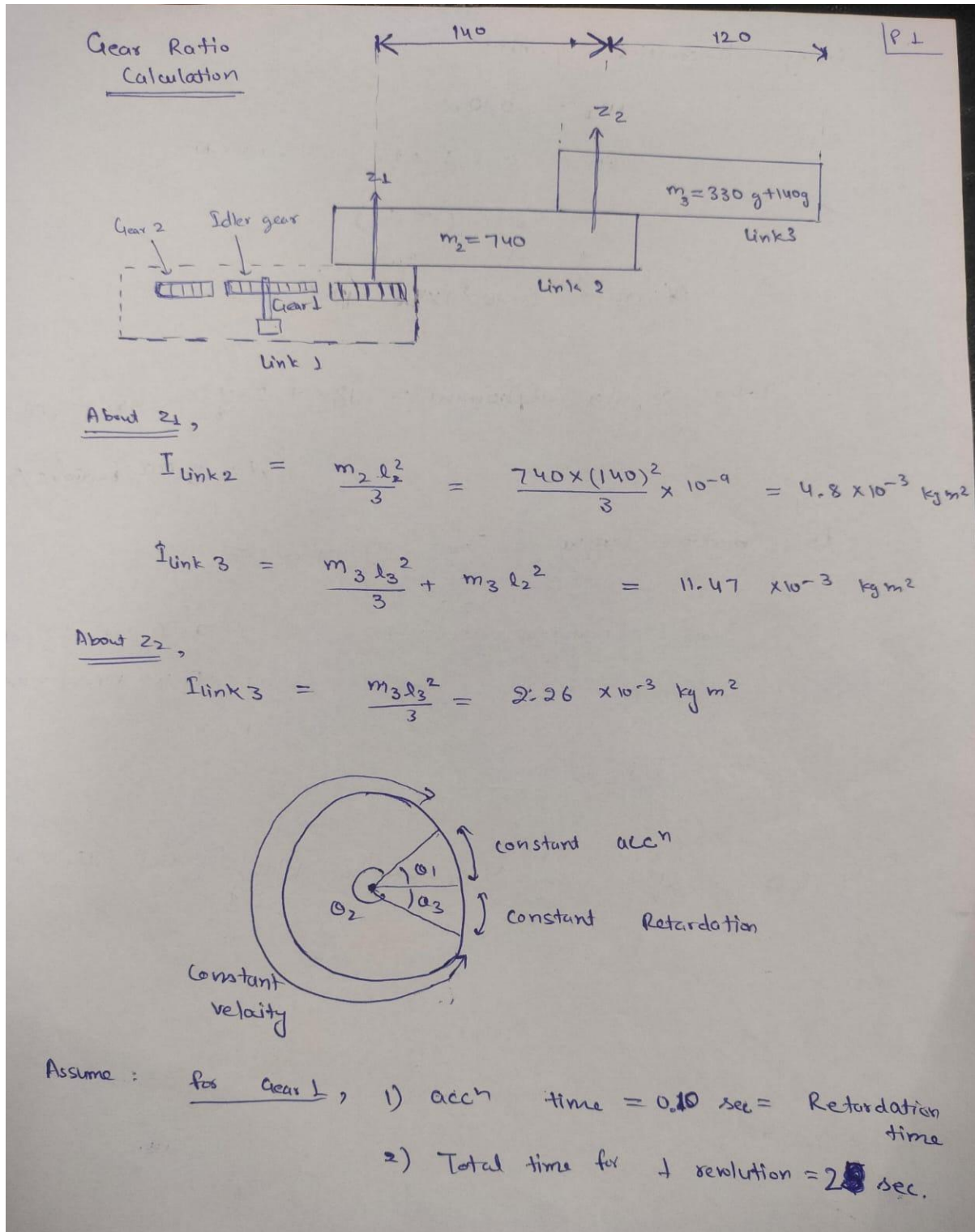


Consider moment due to m_2 and m_3 , acting half on the Bearing and half as force on the collar.

$$\therefore F_{\text{collar}} \times 45 \text{ mm} + M_{\text{bearing}} = M_{\text{due to } m_2 \text{ \& } m_3}$$
$$\Rightarrow 14.1 \times 0.045 + 0.635 = 1.27 \text{ Nm}$$

DYNAMIC ANALYSIS :

Dynamic analysis was performed to study the motion characteristics and inertia effects during robotic operation.



Using kinematics equations!

2P.2

$$\omega_1 = 0.10 \alpha$$

$$\Theta_1 = 0 + \frac{1}{2} \alpha t^2 \quad t = 0.10$$

$$\Theta_1 = \alpha / 800 \quad (= \Theta_3)$$

$$\Theta_2 = (0.10 \alpha) \times 1.8 = 0.18 \alpha$$

$$\therefore \text{Total angular displacement} = 0.18 \alpha + 2 \times \frac{\alpha}{800} = 360^\circ = 2\pi \text{ rad}$$

$$\alpha = 33.04 \text{ radians/s}^2$$

$$\text{As, motor Torque} = T_m = 2 \text{ kg cm} \approx 0.2 \text{ Nm}$$

$$\begin{aligned} \text{Torque required at Gear} &= \alpha (I_{\text{link3}, Z1} + I_{\text{link2}, Z1}) \\ &= 33.04 (11.47 + 4.8) \times 10^{-3} \\ &= 0.5 \text{ Nm} \end{aligned}$$

$$\text{Gear Ratio} = \frac{T_{\text{out}}}{T_{\text{in}}} = \frac{0.5}{0.2} = 2.5$$

Taking Safety margin, we are taking Gear ratio = 2.6

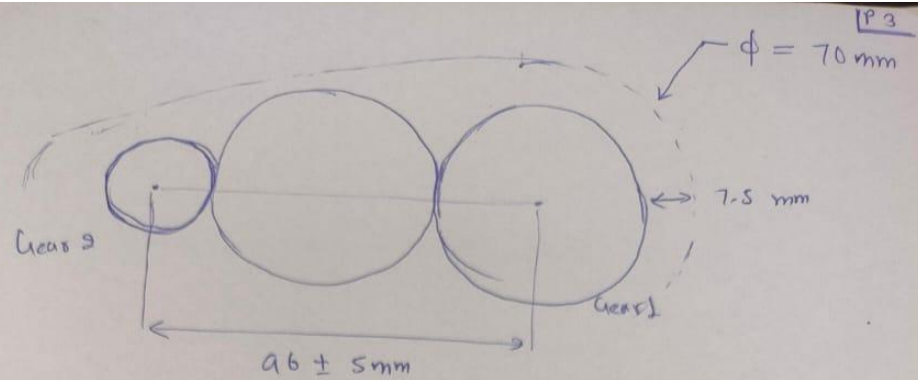
Gear Ratio Calculation

- The gear ratio calculation was performed to determine the torque requirement for smooth motion of the SCARA robotic arm.
- The masses and lengths of Link 2 and Link 3 were considered for rotational inertia calculations.
- Moments of inertia were calculated about axes Z1 and Z2.
- The motion profile of the robotic arm was assumed to consist of:
 - Constant acceleration
 - Constant velocity
 - Constant retardation

- A total time of 2 seconds was assumed for one complete revolution.
- The obtained inertia values were used for further torque and gear ratio calculations.

Kinematic Equations and Gear Ratio

- Kinematic equations were used to determine the angular acceleration of the robotic arm.
- Angular displacement during acceleration and constant velocity motion was calculated.
- The total angular displacement was considered equal to one complete revolution.
- Motor torque provided by the selected motor was compared with the required output torque.
- Gear ratio was calculated based on torque transmission requirements.
- A safety margin was considered while selecting the final gear ratio.
- Final gear ratio selected for the mechanism was 2.6.



Considering the dimensions of base container (70 mm) with some clearance, & taking into account availability of standard gear.

$$\therefore D_{\text{gear 1}} = 55.2 \text{ mm} \quad (\text{pitch circle dia})$$

$$\therefore \text{Gear Ratio} = 2.6 = \frac{D_{\text{Gear 1}}}{D_{\text{Gear 2}}}$$

$$\Rightarrow D_{\text{Gear 2}} = 21.2 \rightarrow \text{also standard gear}$$

$$\therefore \text{Average distance b/w gears} = 96 \text{ mm}$$

$$\Rightarrow 96 \text{ mm} = \frac{D_{\text{gear 1}} + D_{\text{gear 2}}}{2} + \text{Didler}$$

$$\Rightarrow \text{Didler} = 57.8 \text{ mm}$$

↓
considering again availability of standard gear
and constraint due to base container

$$\Rightarrow \boxed{\text{Didler} \approx 55.2 \text{ mm}}$$

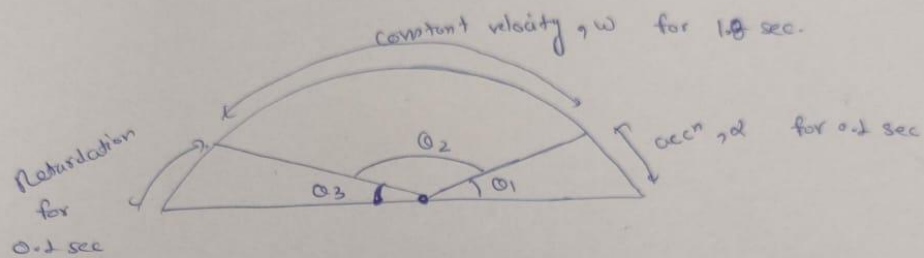
$$\begin{aligned} \therefore \text{Now, Centre to Centre distance b/w gears + 2 gear 2} &= \\ &= 55.2 \times 1.5 + \frac{21.2}{2} \\ &= 93.4 \in [91, 101] \end{aligned}$$

* Gear at junction of Link 2 & Link 3

P-4

About Z_2 ,
$$I_{\text{Link 3}} = \frac{m_3 l_3^2}{3} = 2.26 \times 10^{-3} \text{ kg m}^2$$

Assume : rotation of Link 3 is 180° as shown, in 2 sec.



$$\theta_1 = \frac{1}{2} \alpha (0.1)^2 = 0.005 \alpha \quad (= \theta_3)$$

$$\theta_2 = (0.1 \alpha) 1.8 = 0.18 \alpha$$

$$\Rightarrow 2 \times 0.005 \alpha + 0.18 \alpha = 360^\circ = 2\pi$$

$$\Rightarrow \boxed{\alpha = 16.5 \text{ rad/s}^2}$$

Torque given by motor = $T_m = 19 \text{ mNm}$

Torque required at Gear = $\alpha (I_{\text{Link 3}}, Z_2)$

$$= 16.5 \times 2.26 \times 10^{-3}$$

$$= 37 \text{ mNm}$$

$$\text{Gear ratio} = \frac{T_{\text{out}}}{T_{\text{in}}} = \frac{37}{19} = 1.95$$

Taking safety margin, we are taking Gear ratio = 2

Gear Dimensions Calculation

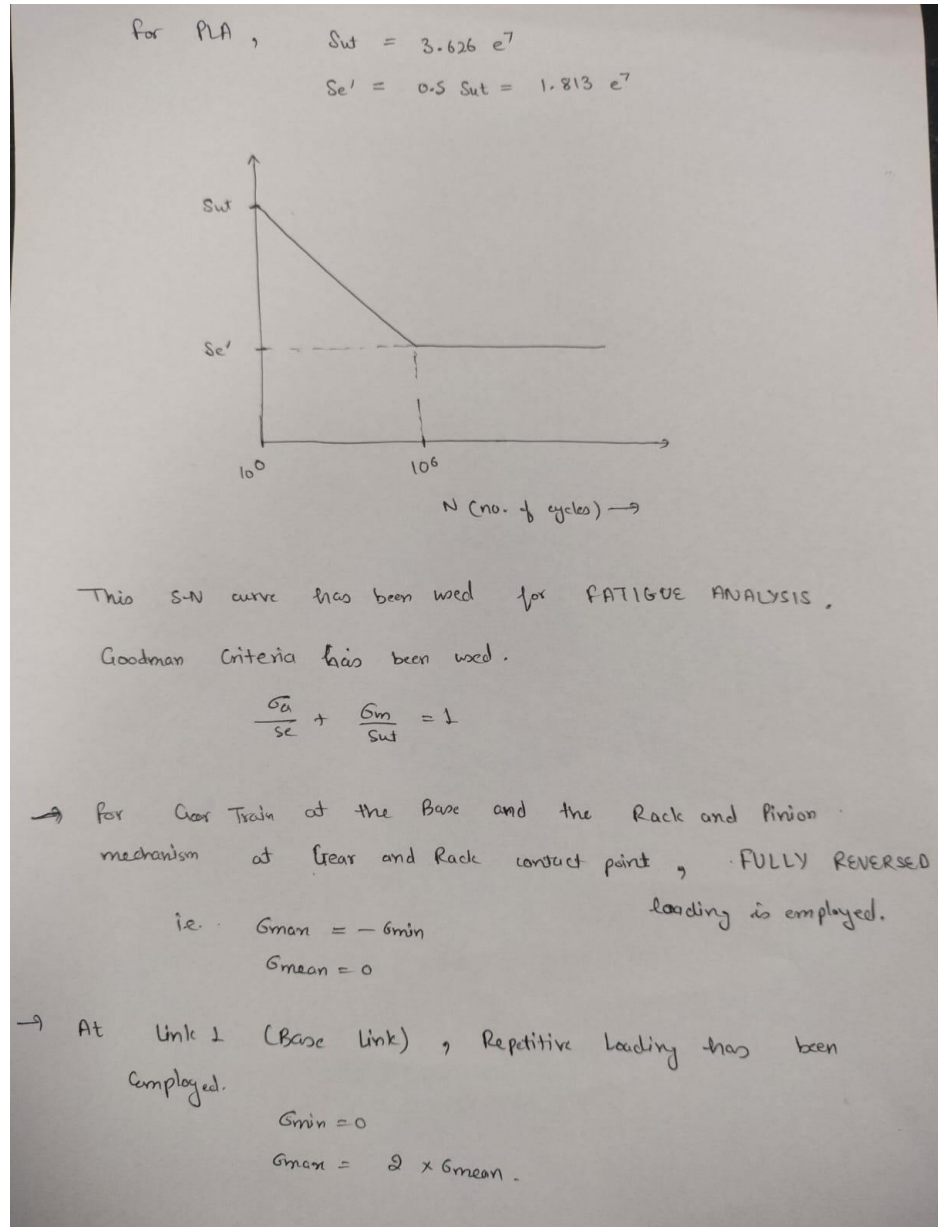
- Gear dimensions were selected considering the space limitations of the base container.
- Standard gear sizes available in the market were considered during design selection.
- Pitch circle diameters of Gear 1 and Gear 2 were calculated using the required gear ratio.
- An idler gear was introduced to maintain proper center distance and motion transfer.
- Clearance between gears was provided to avoid interference during operation.
- Final gear dimensions were selected based on:
 - Gear availability
 - Compact design
 - Proper center distance
 - Smooth power transmission

Gear Calculation at Junction of Link 2 and Link 3

- Gear calculations were performed at the junction of Link 2 and Link 3 to determine torque transmission requirements.
- The moment of inertia of Link 3 about axis Z_{2Z2} was calculated.
- Motion of Link 3 was assumed to be 180° rotation within 2 seconds.
- The motion profile included acceleration, constant velocity, and retardation regions.
- Angular acceleration was determined using kinematic relations.
- Required output torque at the gear was calculated using rotational inertia and angular acceleration.
- Gear ratio was obtained by comparing required torque and motor torque.
- A safety margin was included while selecting the final gear ratio.
- Final gear ratio selected at the junction was 2.0.

Fatigue Analysis Using S-N Curve

Fatigue analysis of the robotic components was performed using the S-N curve for PLA material to study the effect of cyclic loading.



- The Goodman failure criterion was used to evaluate fatigue safety under fluctuating stresses.
- Fully reversed loading condition was considered for the gear train and rack-and-pinion contact region, where tensile and compressive stresses alternate continuously.
- Repetitive loading condition was considered for Link 1 (base link), where stress varies from zero to maximum value during operation.
- The fatigue analysis helped in predicting the durability and safe operating life of the robotic mechanism under repeated loading conditions.

A: Static Structural

Equivalent Stress

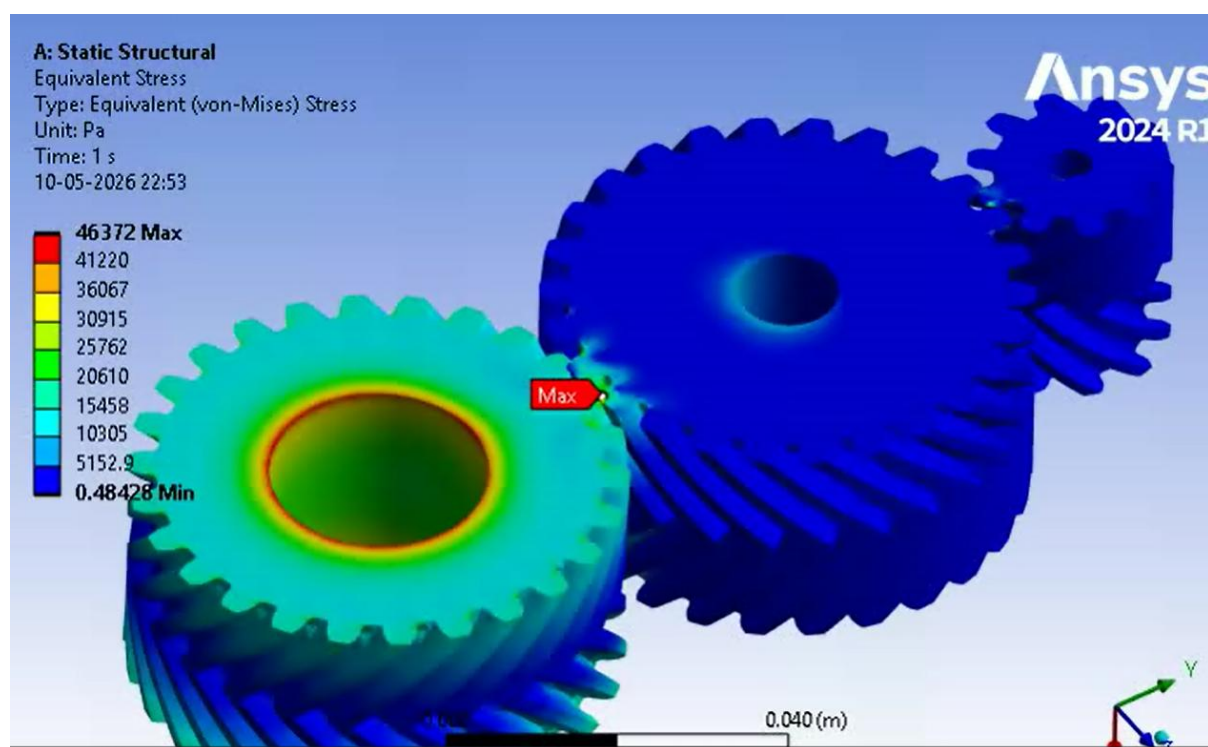
Type: Equivalent (von-Mises) Stress

Unit: Pa

Time: 1 s

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Ansys
2024 R1



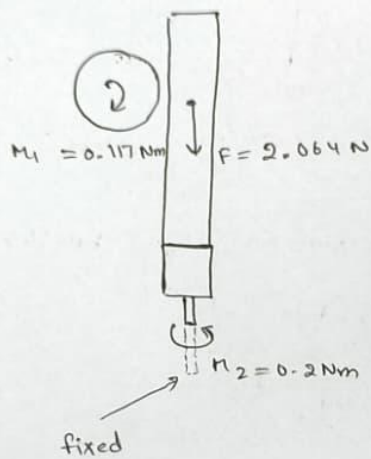
For static loading

FOS base link = 4.8

For fatigue loading

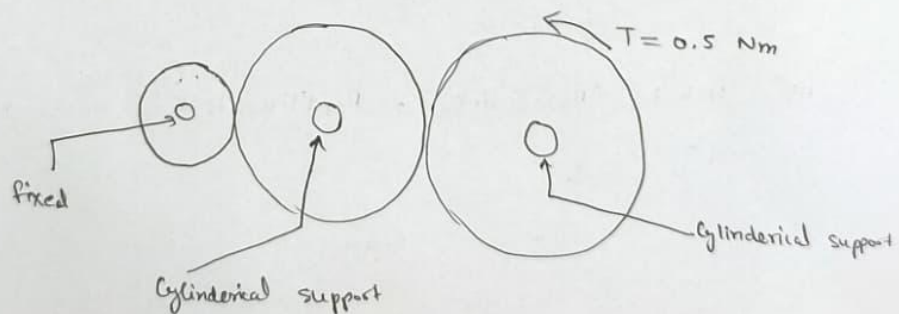
FOS base link = 4

* Boundary Conditions used for FEM:
for rack and pinion:



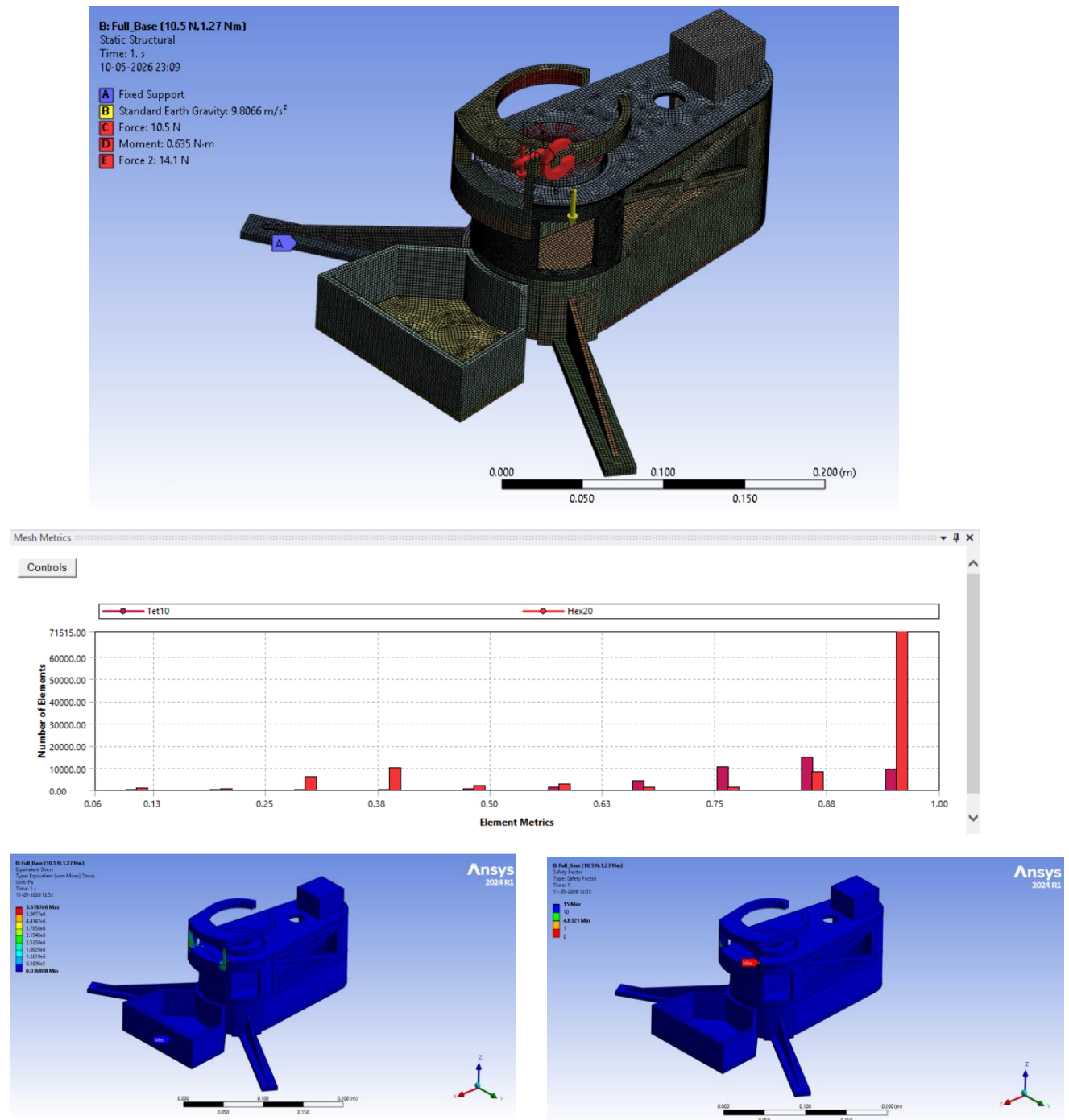
No occurrence of failure
as
magnitude of loadings
is very less

For Gear Train (at Base)

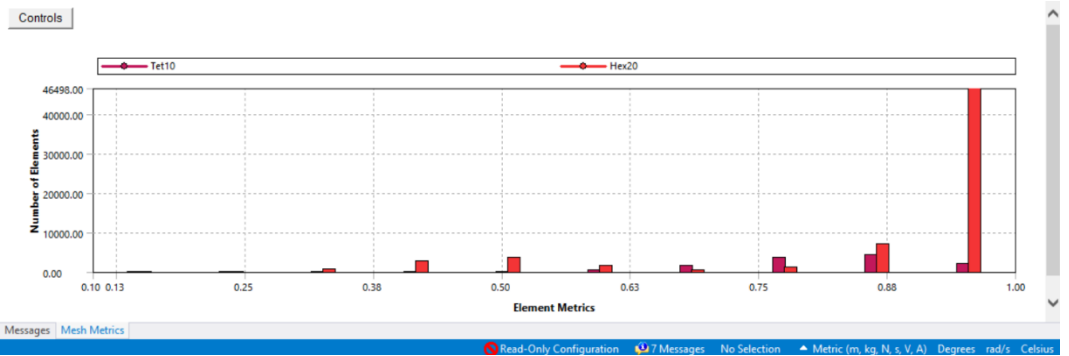
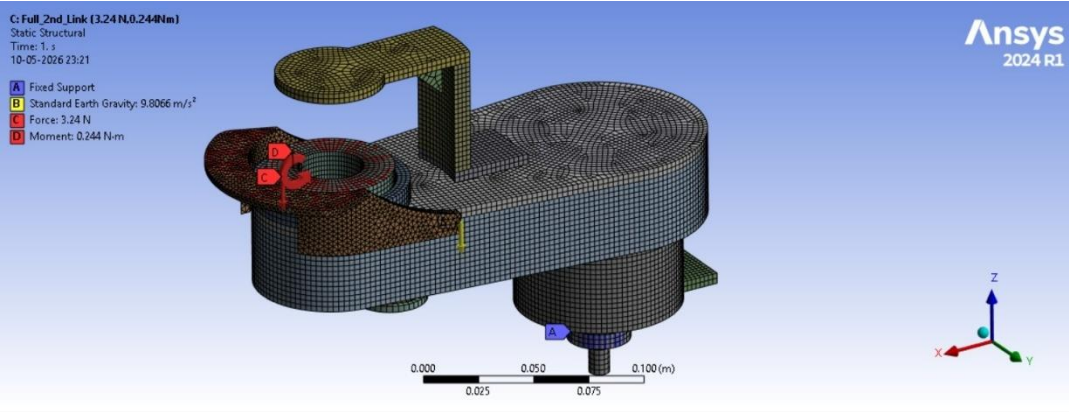


Meshing (Structured grid generation)

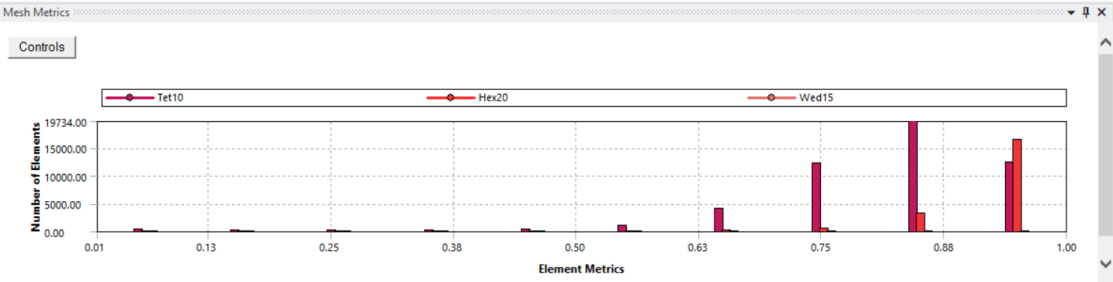
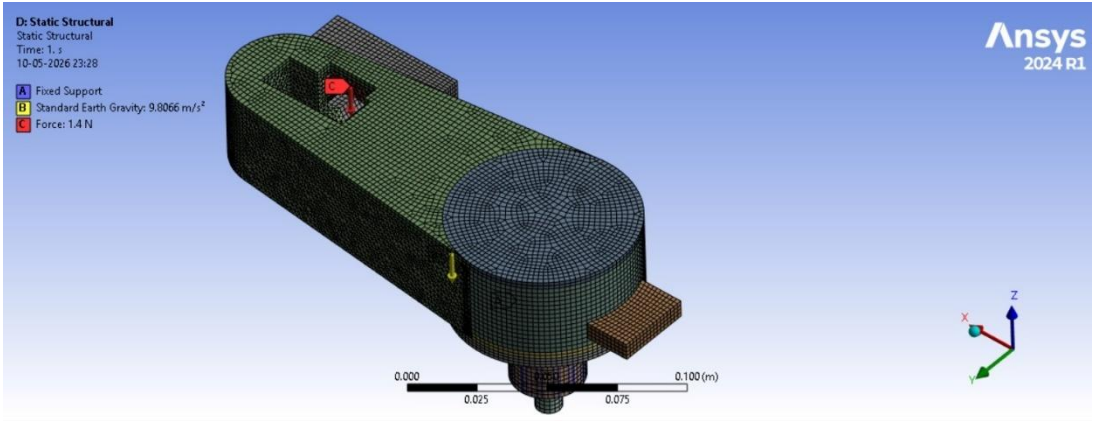
Base Link



Second link



Third Link



Rack and Pinion

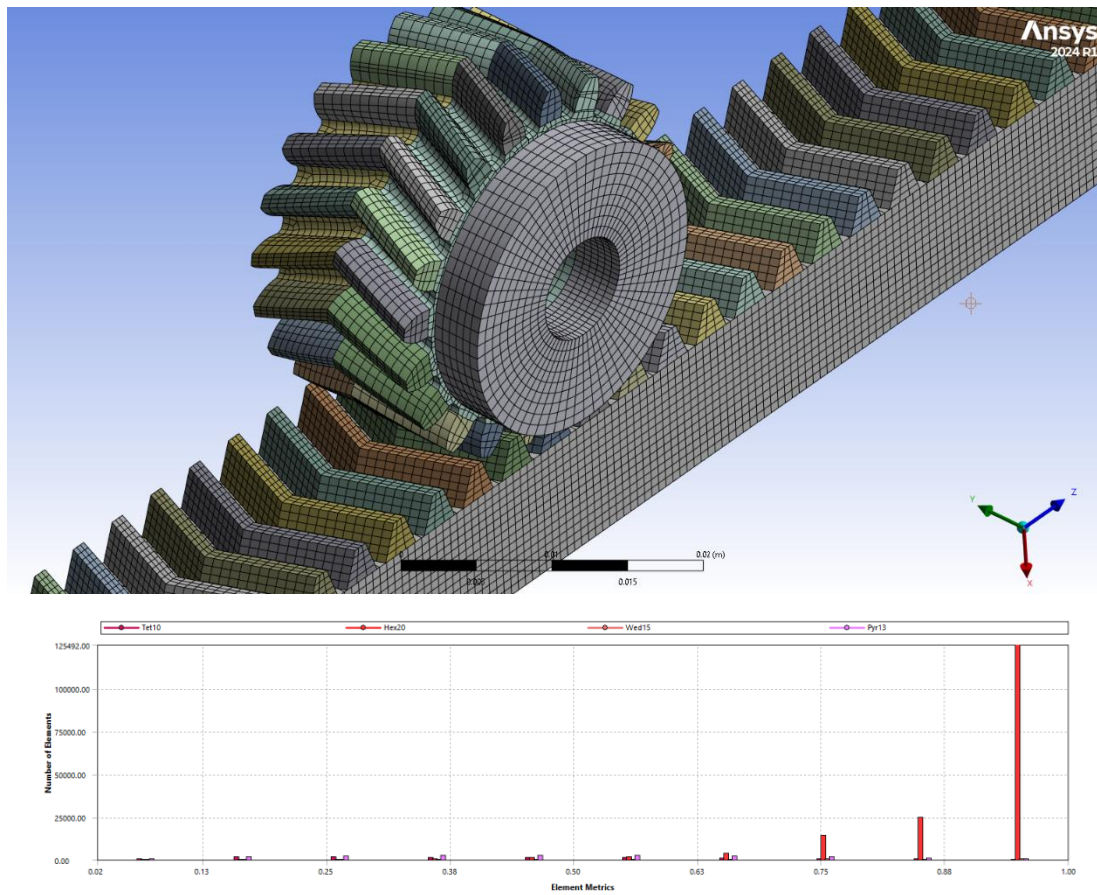


Fig. Mesh Metrics for meshing of Rack and pinion

No of nodes= 841065

No of elements = 196168

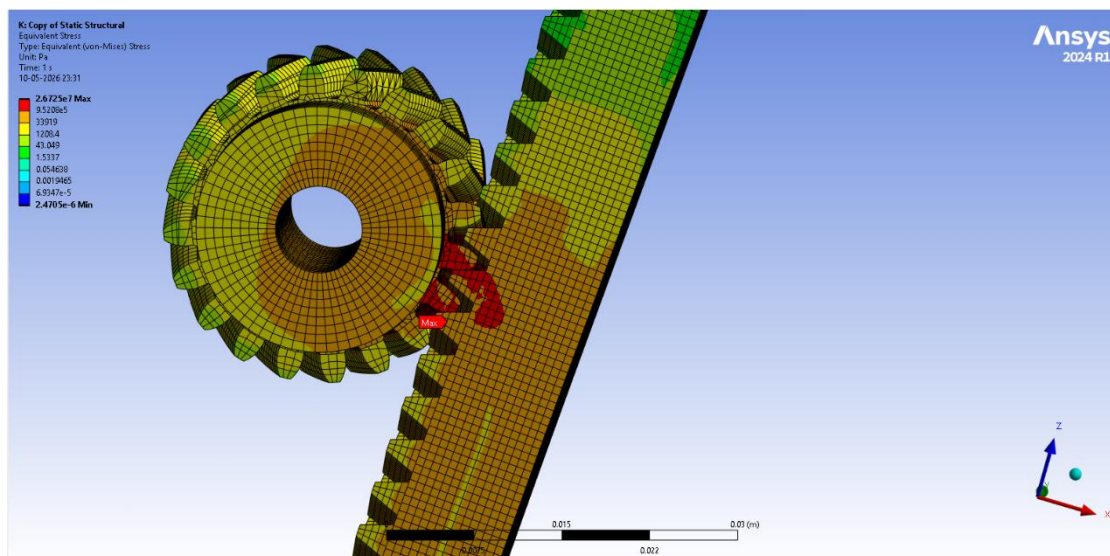
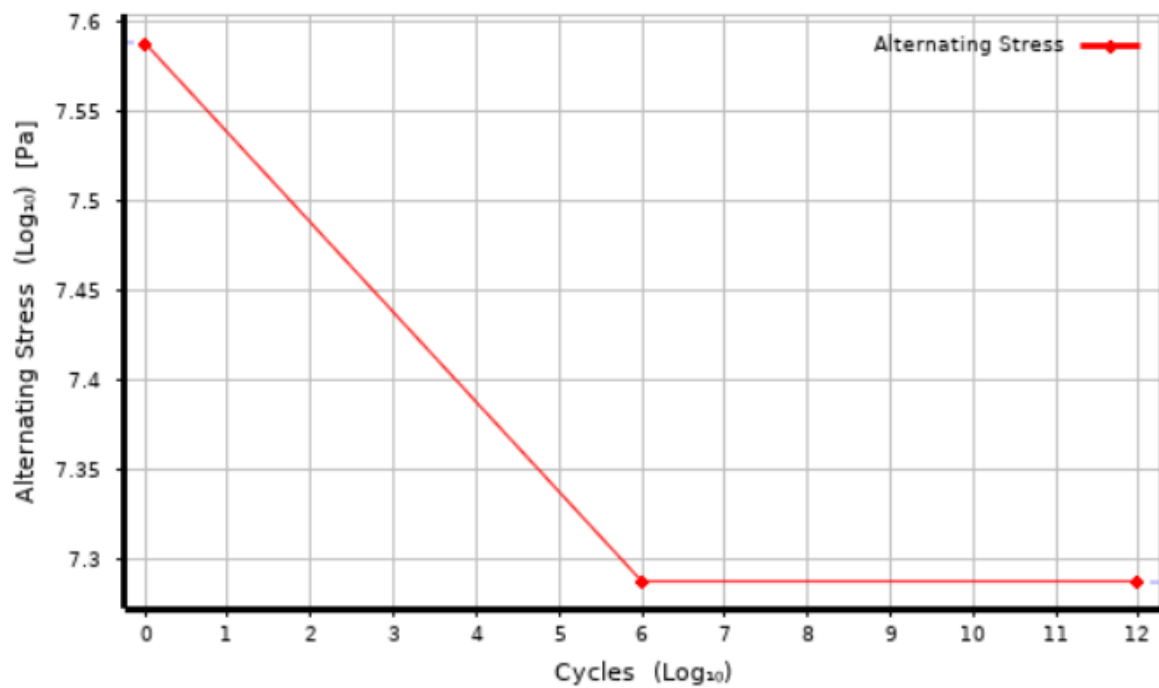


Table of Properties Row 15: S-N Curve

	A	B
1	Cycles 	Alternating Stress (Pa) 
2	1	3.873E+07
3	1E+06	1.9365E+07
4	1E+12	1.9365E+07
*		

Chart of Properties Row 15: S-N Curve



HARDWARE COMPONENTS

Hardware Component	Function	Reason for Selection
Arduino Uno	Main controller of the robotic system	Easy programming and compatibility with drivers and sensors
TB6600 Stepper Motor Driver	Drives NEMA 17 stepper motor	Provides sufficient current and better thermal performance
NEMA 17 Stepper Motor	Primary rotational motion of robotic arm	High torque and precise positioning
28BYJ-48 Stepper Motor	Lightweight rotational motion	Compact size and low power consumption
ULN2003 Driver	Drives 28BYJ-48 motor	Current amplification for motor operation
N20 Geared Motor	Low speed high torque motion	Compact design with gearbox for controlled motion
Limit Switch	Motion limit detection and homing	Prevents overtravel and improves safety

Driver Selection Observation

Driver Used	Observation	Result
DRV8825	Heating at around 1.5 A current	Current reduced to 0.6 A to reduce heating
DRV8825 at 0.6 A	Torque became insufficient	Unable to operate robotic arm smoothly
TB6600 Driver	Higher current capacity with less heating	Provided sufficient torque and stable operation

Arduino Code

```
// =====  
// SCARA ROBOT - FULL MANUAL CONTROL  
//  
// LINK 1 -> NEMA17 + TB6600  
// LINK 2 -> 28BYJ-48 + ULN2003  
// LINK 3 -> 28BYJ-48 + ULN2003 + DC MOTOR  
//  
// FEATURES:  
// 1. Move Link 1 by steps (+ or -)  
// 2. Move Link 2 by steps (+ or -)  
// 3. Calibrate Link 3  
// 4. Link 3 Forward Sequence  
// 5. Link 3 Reverse Sequence  
// =====  
  
#include <AccelStepper.h>  
  
// =====  
// LINK 1 -> NEMA17 + TB6600  
// =====  
#define J1_STEP_PIN 3  
#define J1_DIR_PIN 2  
#define J1_LIMIT_PIN A3  
  
// =====  
// LINK 2 -> 28BYJ-48 + ULN2003  
// =====  
#define J2_IN1 8  
#define J2_IN2 10  
#define J2_IN3 9  
#define J2_IN4 11  
#define J2_LIMIT_PIN 13  
  
// =====  
// LINK 3 -> 28BYJ-48 + ULN2003  
// =====  
#define J3_IN1 4  
#define J3_IN2 6  
#define J3_IN3 5  
#define J3_IN4 7  
#define J3_LIMIT_PIN 12
```

```

// =====
// DC MOTOR PINS
// =====
#define ENA A0
#define INA1 A1
#define INA2 A2

// =====
// MOTOR TYPES
// =====
#define DRIVER_INTERFACE 1
#define FULL4WIRE 4

// =====
// MOTOR OBJECTS
// =====
AccelStepper joint1(DRIVER_INTERFACE,
    J1_STEP_PIN,
    J1_DIR_PIN);

AccelStepper joint2(FULL4WIRE,
    J2_IN1,
    J2_IN2,
    J2_IN3,
    J2_IN4);

AccelStepper joint3(FULL4WIRE,
    J3_IN1,
    J3_IN2,
    J3_IN3,
    J3_IN4);

// =====
// VARIABLES
// =====
String command;
long steps;

// =====
// LIMIT SWITCH FUNCTION
// =====
bool limitTriggered(int pin)
{
    return (digitalRead(pin) == LOW);
}

// =====
// DC MOTOR FUNCTIONS

```

```

// =====
void stopDCMotor()
{
    digitalWrite(INA1, LOW);
    digitalWrite(INA2, LOW);

    analogWrite(ENA, 0);
}

void rotateDCMotorForward(int speedValue)
{
    digitalWrite(INA1, HIGH);
    digitalWrite(INA2, LOW);

    analogWrite(ENA, speedValue);
}

void rotateDCMotorBackward(int speedValue)
{
    digitalWrite(INA1, LOW);
    digitalWrite(INA2, HIGH);

    analogWrite(ENA, speedValue);
}

// =====
// LINK 3 CALIBRATION
// =====
void calibrateLink3()
{
    Serial.println("=====");
    Serial.println("LINK 3 CALIBRATION STARTED");
    Serial.println("=====");

    joint3.setMaxSpeed(300);
    joint3.setAcceleration(100);

    // Move toward limit switch
    joint3.moveTo(-100000);

    while (!limitTriggered(J3_LIMIT_PIN))
    {
        joint3.run();
    }

    Serial.println("Limit Switch Triggered");

    joint3.stop();
}

```

```

joint3.runToPosition();

joint3.setCurrentPosition(0);

delay(500);

// Move slightly away from switch
Serial.println("Moving to Home Position");

joint3.move(50);

while (joint3.distanceToGo() != 0)
{
    joint3.run();
}

joint3.setCurrentPosition(0);

Serial.println("Calibration Complete");
}

// =====
// MOVE LINK 1
// =====
void moveLink1(long targetSteps)
{
    Serial.print("Moving Link 1 Steps = ");
    Serial.println(targetSteps);

    joint1.setMaxSpeed(800);
    joint1.setAcceleration(500);

    // Positive -> Forward
    // Negative -> Reverse
    joint1.move(targetSteps);

    while (joint1.distanceToGo() != 0)
    {
        joint1.run();
    }

    Serial.println("Link 1 Motion Complete");
}

// =====
// MOVE LINK 2
// =====
void moveLink2(long targetSteps)

```

```

{
  Serial.print("Moving Link 2 Steps = ");
  Serial.println(targetSteps);

  joint2.setMaxSpeed(500);
  joint2.setAcceleration(300);

  // Positive -> Forward
  // Negative -> Reverse
  joint2.move(targetSteps);

  while (joint2.distanceToGo() != 0)
  {
    joint2.run();
  }

  Serial.println("Link 2 Motion Complete");
}

// =====
// LINK 3 FORWARD SEQUENCE
// =====
void forwardSequence()
{
  Serial.println("=====");
  Serial.println("FORWARD SEQUENCE STARTED");
  Serial.println("=====");

  // =====
  // FAST MOVE
  // =====
  joint3.setMaxSpeed(600);
  joint3.setAcceleration(300);

  joint3.move(1200);

  while (joint3.distanceToGo() != 0)
  {
    joint3.run();
  }

  Serial.println("1200 Steps Reached");

  // =====
  // START DC MOTOR FORWARD
  // =====
  Serial.println("Starting DC Motor Forward");
}

```

```

rotateDCMotorForward(80);

delay(2000);

// =====
// SLOW MOVE
// =====
joint3.setMaxSpeed(250);
joint3.setAcceleration(200);

joint3.move(700);

while (joint3.distanceToGo() != 0)
{
    joint3.run();
}

// =====
// STOP EVERYTHING
// =====
stopDCMotor();

Serial.println("Forward Sequence Complete");
}

// =====
// LINK 3 REVERSE SEQUENCE
// =====
void reverseSequence()
{
    Serial.println("=====");
    Serial.println("REVERSE SEQUENCE STARTED");
    Serial.println("=====");

    // =====
    // START DC MOTOR REVERSE
    // =====
    rotateDCMotorBackward(80);

    delay(1000);

    // =====
    // REVERSE STEPPER
    // =====
    joint3.setMaxSpeed(600);
    joint3.setAcceleration(300);

    joint3.move(-1000);

```

```

while (joint3.distanceToGo() != 0)
{
    joint3.run();
}

// =====
// STOP EVERYTHING
// =====
stopDCMotor();

Serial.println("Reverse Sequence Complete");
}

// =====
// SETUP
// =====
void setup()
{
    Serial.begin(9600);

    // =====
    // LIMIT SWITCH PINS
    // =====
    pinMode(J1_LIMIT_PIN, INPUT_PULLUP);
    pinMode(J2_LIMIT_PIN, INPUT_PULLUP);
    pinMode(J3_LIMIT_PIN, INPUT_PULLUP);

    // =====
    // DC MOTOR PINS
    // =====
    pinMode(ENA, OUTPUT);
    pinMode(INA1, OUTPUT);
    pinMode(INA2, OUTPUT);

    stopDCMotor();

    Serial.println("=====");
    Serial.println("SCARA ROBOT READY");
    Serial.println("=====");
}

// =====
// LOOP
// =====
void loop()
{
    Serial.println();
}

```



```

Serial.println("=====");
Serial.println("ENTER COMMAND:");
Serial.println("1 = Move Link 1");
Serial.println("2 = Move Link 2");
Serial.println("3 = Calibrate Link 3");
Serial.println("f = Link 3 Forward Sequence");
Serial.println("r = Link 3 Reverse Sequence");
Serial.println("=====");

// Wait for user input
while (Serial.available() == 0)
{
}

command = Serial.readStringUntil('\n');

command.trim();

// =====
// LINK 1 CONTROL
// =====
if (command == "1")
{
    Serial.println("Enter Steps for Link 1:");
    Serial.println("Positive = Forward");
    Serial.println("Negative = Reverse");

    while (Serial.available() == 0)
    {
    }

    steps = Serial.parseInt();

    while (Serial.available())
    {
        Serial.read();
    }

    moveLink1(steps);
}

// =====
// LINK 2 CONTROL
// =====
else if (command == "2")
{
    Serial.println("Enter Steps for Link 2:");
    Serial.println("Positive = Forward");

```

```

Serial.println("Negative = Reverse");

while (Serial.available() == 0)
{
}

steps = Serial.parseInt();

while (Serial.available())
{
    Serial.read();
}

moveLink2(steps);
}

// =====
// LINK 3 CALIBRATION
// =====
else if (command == "3")
{
    calibrateLink3();
}

// =====
// LINK 3 FORWARD
// =====
else if (command == "f")
{
    forwardSequence();
}

// =====
// LINK 3 REVERSE
// =====
else if (command == "r")
{
    reverseSequence();
}

// =====
// INVALID COMMAND
// =====
else
{
    Serial.println("Invalid Command");
}

```

```
Serial.println();  
Serial.println("NEXT COMMAND READY");
```

} FUTURE SCOPE

Future improvements for the project include:

- Integration of machine vision systems
- IoT-based monitoring and control
- Artificial intelligence for autonomous operation
- Increased payload capacity
- Wireless control system
- Advanced motion planning algorithms

These developments can improve industrial applicability and operational efficiency.

CONCLUSION

The project successfully demonstrated the design, analysis, and development of a SCARA robotic mechanism for industrial automation applications. CAD modeling, force analysis, static analysis, dynamic analysis, bending moment calculations, and fatigue analysis were performed to ensure structural safety and efficient operation.

The developed robotic system provides advantages such as compact design, reduced weight, improved motion accuracy, and low operational cost. The obtained factor of safety confirms that the structure can safely withstand the applied loading conditions.

The project provides practical understanding of robotic mechanism design, stress analysis, and system integration for industrial automation applications.

REFERENCES

[Video](https://www.youtube.com/watch?v=1QHJksTrk8s) which is taken as reference (<https://www.youtube.com/watch?v=1QHJksTrk8s>)